COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 7 — Agile Model

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Section 11: Classroom Activity — Prioritize Features for Technical Festival

Scenario: The annual college tech festival occurs in 3 weeks. The team has limited developer bandwidth and must prioritize 5 candidate features: Event Listings, Student Registration, QR Code Gate Attendance, Digital Certificate Generator, and Advanced Analytics Dashboard.

1. Feature Backlog Prioritization (MoSCoW Framework)

Priority Tier	Feature	Complexity / Effort	Strategic Rationale
Priority 1 (Must Have)	Event Listings & Details	Low (3 Story Points)	Absolute baseline prerequisite. Students cannot register without knowing schedules, rules, venues, and timings.
Priority 2 (Must Have)	Student Online Registration	Medium (8 Story Points)	Core transactional requirement. Enforces capacity limits and generates verified attendee lists needed by event heads.
Priority 3 (Should Have)	QR Code Gate Attendance	Medium (5 Story Points)	Prevents bottleneck queues at venue gates. If time runs short, paper check-off acts as manual fallback.
Priority 4 (Could Have)	Automated Certificate Generator	Medium (5 Story Points)	Valuable but non-urgent. Certificates are distributed after the festival concludes; can be built during post-fest sprint.
Priority 5 (Won't Have)	Advanced Analytics Dashboard	High (13 Story Points)	Complex data aggregation. Adds zero immediate value to student attendees or gate volunteers during the 3-week sprint.

2. Sprint 1 Delivery Scope

The team will commit strictly to Priority 1 (Event Listings) and Priority 2 (Student Online Registration) in Sprint 1 (Days 1–10). This establishes a functioning Minimum Viable Product (MVP) that allows the college to open registrations immediately.

Section 14: Homework — Agile Plan for a Student Portal

- **1. One-Sentence Product Goal:** "Empower university students with a centralized, mobile-responsive portal to track real-time course grades, manage semester registrations, and receive urgent campus academic alerts."
- **2. Prioritized Feature Backlog (Ranked 1 to 6):** (1) Student Profile & Course Enrollment View; (2) Internal Assessment & Exam Grade Tracker; (3) Fee Payment & Digital Receipt Download; (4) Academic Timetable & Class Cancellation Alerts; (5) Faculty Appointment Booking; (6) Student Club & Extracurricular Directory.
- **3. Justification for First Two Features:** Course enrollment viewing and grade tracking solve the most frequent, high-anxiety student inquiries and form the core database entity model required for all subsequent portal features.
- **4. Post-Release Feedback Questions:** (a) "Did your current semester registered subjects display correctly?"; (b) "How easily did you find your internal assessment marks on mobile?"; (c) "What single feature did you expect to see that was missing?"
- **5. Identified Project Risk & Agile Investigation:** Risk: Legacy college ERP database uses archaic SOAP endpoints with high latency. Investigation: Implement a time-boxed 2-day developer spike in Sprint 0 to test caching API responses in Redis.

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